

Industrial Machine Failure Prediction

Learn Depth Academy LLP — Track 1 Final Capstone — Problem 21

Prepared as a capstone project starter / technical report

1. Executive Summary

This project develops an end-to-end machine-learning prototype for predicting whether an industrial machine is likely to experience failure within a defined operating window. The selected problem is Problem 21 in the supplied Learn Depth Academy capstone brief. The solution uses an openly available predictive-maintenance dataset, performs data preparation and exploratory analysis, trains a foundational classifier, evaluates classification performance, and exposes the model through a lightweight Streamlit application.

The capstone brief explicitly asks for free/open-source tools, foundational models such as Logistic Regression, KNN or Decision Tree, interpretation of Accuracy/Precision/Recall/F1-score and a confusion matrix, and a Streamlit prototype.

2. Problem Definition

The supplied brief defines the task as binary classification in the Industrial Systems domain: predict whether a machine is likely to experience a failure within a defined operating window. The expected input signals include temperature, vibration, pressure, runtime and maintenance-gap information.

3. Dataset Selection

The implementation uses the AI4I 2020 Predictive Maintenance Dataset from the UCI Machine Learning Repository. UCI describes it as a synthetic dataset reflecting predictive-maintenance data encountered in industry, with 10,000 observations and variables including air temperature, process temperature, rotational speed, torque, tool wear and machine-failure status. The dataset is licensed under CC BY 4.0.

Source: UCI Machine Learning Repository, AI4I 2020 Predictive Maintenance Dataset, DOI 10.24432/C5HS5C.

4. Objectives

1	Understand the factors associated with machine failure.
2	Prepare and inspect the dataset for quality, missing values, duplicates, imbalance and leakage.
3	Train a reproducible foundational binary classifier.
4	Evaluate the model using classification metrics and a confusion matrix.
5	Build a lightweight Streamlit prediction interface.
6	Document limitations, future scope and references.

5. Methodology

Data loading: The Python implementation fetches the UCI dataset using the ucimlrepo package. Identifier columns are removed from the predictive feature set. The target is machine failure.

Preprocessing: Numeric variables are median-imputed and standardized. The categorical product type is most-frequent imputed and one-hot encoded. The preprocessing is placed inside a scikit-learn Pipeline so the same transformations are used during training and prediction.

Model: Logistic Regression with class balancing is used as the initial foundational model. The class-weight setting is included because failure cases are relatively uncommon in this dataset.

Evaluation: Accuracy, Precision, Recall, F1-score, ROC-AUC and a confusion matrix are generated. The interpretation should focus especially on false negatives because missing a potential failure can be more consequential than producing an extra alert in an educational predictive-maintenance scenario.

6. EDA and Investigation Plan

Area	What to investigate
Data quality	Missing values, duplicate records, data types and impossible values
Class balance	Frequency of failure vs non-failure observations
Distributions	Temperature, speed, torque and tool-wear distributions
Relationships	Feature relationships with the machine-failure target
Outliers	Extreme operating conditions and whether they correspond to failures
Leakage	Any feature that directly or indirectly reveals the target
Error analysis	False positives and false negatives after model evaluation

7. Results Status

The supplied capstone document requires reproducible results and states that students must be able to explain and defend their implementation. Therefore, this report does **not** invent final accuracy or F1 values. The included training script generates the real evaluation metrics after the UCI dataset is downloaded and the pipeline is executed.

8. Prototype Design

The Streamlit application provides inputs for product type, air temperature, process temperature, rotational speed, torque and tool wear. It loads the same serialized pipeline used during training and returns the predicted failure class and estimated probability. The application is explicitly presented as an educational prototype, not a production safety system.

9. Limitations

The UCI dataset is synthetic rather than a live industrial plant dataset. The capstone prototype is therefore suitable for demonstrating the ML workflow but should not be interpreted as a validated industrial safety system. A real deployment would require plant-specific sensor data, temporal validation, monitoring, calibration, drift handling and domain validation.

10. Future Scope

Future work can compare Logistic Regression with Decision Tree and KNN, investigate threshold tuning for the failure class, add explainability, test temporal or grouped validation strategies, and integrate maintenance-history features when a suitable open dataset is available.

11. 10-Day Work Plan

Day 1	Problem definition + research
Day 2	Dataset acquisition + data understanding
Day 3	EDA + data quality
Day 4	Baseline model
Day 5	Evaluation
Day 6	Improvement / threshold and feature checks
Day 7	Streamlit prototype
Day 8	Integration + testing
Day 9	Technical paper + presentation
Day 10	Demo + viva preparation

12. References

1. UCI Machine Learning Repository. AI4I 2020 Predictive Maintenance Dataset. DOI: 10.24432/C5HS5C.

2. Learn Depth Academy LLP. Track 1 Final Capstone Problem 21: Industrial Machine Failure Prediction (supplied project brief).

Note: This document is a project starter report. Replace the Results Status section with the actual metrics, plots and interpretation produced by the supplied training pipeline before final submission.